

Preparation and Performance of Castor Oil Polyurethane Marine Antifouling Coatings with Acrylamide Quaternary Ammonium Salts as Antimicrobial Agents

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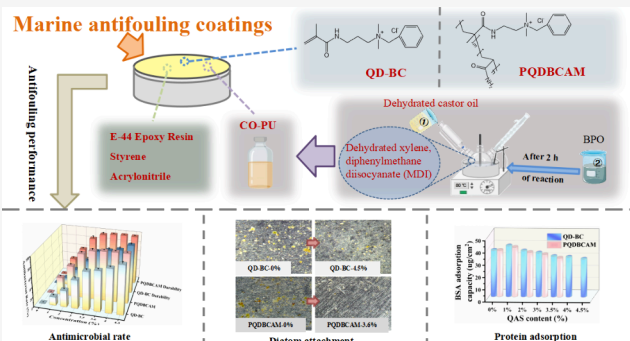
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ABSTRACT: The surfaces of underwater ship hulls and aquaculture equipment, such as fish cages, are highly susceptible to damage from fouling organisms. Although traditional marine antifouling coatings exhibit effective antifouling properties, the leaching of antifouling agents into the marine environment can lead to pollution and ecological disruption. In this study, we prepared castor oil polyurethane (CO-PU) by reacting castor oil with isocyanate. We then incorporated self-synthesized acrylamide-based quaternary ammonium salts (QASs), specifically dimethyloctylaminopropyl methacrylamide-ammonium QD-BC and its polymer PQDBCAM, into the CO-PU resin to develop CO-PU marine antifouling coatings. By optimizing the formulation to enhance the cross-linking degree of the coating, we obtained coatings with improved mechanical properties and antifouling performance. The results indicate that, in comparison to the pure CO-PU coating, the hydrophilicity of the coating is enhanced, the flexibility is superior, the pencil hardness increases from 5H to 6H, and the adhesion of the PQDBCAM antifouling coating reaches a maximum of 4.79 MPa. All of the coatings demonstrated effectiveness in inhibiting the growth of *Pseudomonas aeruginosa*, diatoms, and protein attachment, and the increase of QASs leads to enhanced effects. This suggests that acrylamide QAS marine antifouling coatings have a certain degree of antifouling performance, and polymer-based quaternary ammonium PQDBCAM antifouling coatings show superior efficacy. After the 3.6% PQDBCAM coating was statically placed in diatoms for 7 days, the coverage area of diatoms was merely approximately 22.3% and the protein adsorption amount on the surface of the antifouling coating was 31.72 $\mu\text{g}/\text{cm}^2$. The coating could maintain its integrity after 3 months and still exhibit excellent antibacterial effects. The antifouling effect was more durable, effectively reducing the maintenance times of ships and the cleaning frequency of aquaculture equipment.



INTRODUCTION

When humans undertake marine operations, it is necessary to take into account the threats and losses caused by marine fouling. Antifouling coatings can effectively prevent the substrate from being contaminated;^{1,2} however, the existing antifouling coatings suffer from shortcomings, such as short service life, poor mechanical properties, high toxicity, and high cost. Against the background of significantly elevated global standards for the safety and environmental friendliness of materials utilized in the marine domain, scholars in this field have conducted a large amount of research, mainly concentrating on seeking more effective and environmentally friendly antibacterial antifouling agents and biomimetic surface microstructures for preventing biofouling, among other areas.

Polyurethane resin is synthesized through the reaction between polyols and isocyanates. Owing to its outstanding mechanical properties and abrasion resistance,^{3–5} polyurethane

resin has been extensively applied in the preparation of various coatings. Consequently, polyurethane resin can be employed as antifouling coatings to safeguard the substrate from contamination.^{6–8} The materials for synthesizing polyurethane resin mainly consist of two categories: polyol compounds and isocyanates. The raw materials for synthesizing polyols mostly originate from the petrochemical industry. For instance, ethylene glycol is obtained by oxidizing ethylene from the petrochemical industry and subsequently undergoing hydrol-

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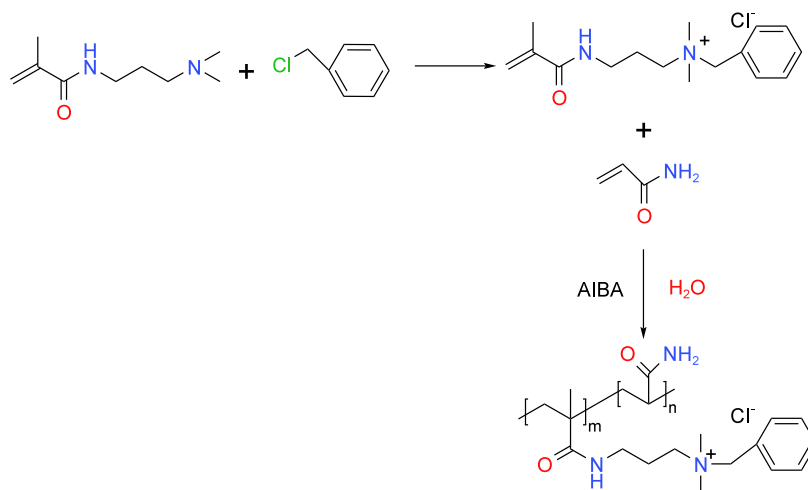


Figure 1. Synthesis of the QASs.

ysis. Nevertheless, petroleum-based raw materials are non-renewable resources, and their reserves are gradually diminishing with continuous extraction. In contrast, vegetable oils possess hydroxyl groups and unsaturated double bonds, which can be utilized as polyols in the manufacturing of polyurethane. This method provides a feasible alternative to the utilization of non-renewable petroleum-based raw materials, thereby reducing the consumption of resources.^{9–12} Patil et al.¹³ fabricated thermoset polyurethane coatings by converting renewable cottonseed oil into polyester amide polyol, which was subsequently reacted with diisocyanate. It was ascertained that the fabricated coatings displayed optimal glossiness, adhesion, pencil hardness, high suitability for the curing process, resistance to both acids and alkalis, and hydrophobic properties.

China is among the world leaders in the production of castor oil and other vegetable oils. Castor oil offers advantages, such as renewability and cost-effectiveness. It is a light yellow, transparent liquid with a molecular structure rich in hydroxyl groups. After modification, castor oil can serve as a bio-based polyol, effectively replacing petroleum-based raw materials and reducing dependence upon non-renewable resources. Polyurethane resins can be synthesized by directly reacting the hydroxyl groups in castor oil with isocyanates.^{14,15} While corn oil, soybean oil, and other vegetable oils are extensively used in the preparation of polyurethane coatings and have significant production volumes, their use must also meet human food and other demands, potentially leading to challenges in ensuring a stable raw material supply. It is feasible to fully utilize this advantageous feature of castor oil to prepare bio-based polyurethane coatings for marine antifouling. Zhou et al.¹⁶ synthesized a series of polyurethane coatings using castor oil as a polyol and reacting it with hexamethylene diisocyanate trimer and monomethanol-terminated polydimethylsiloxane. The resultant coatings exhibited a pencil hardness of up to 6H and demonstrated outstanding antifouling and anticorrosion properties. Bin et al.¹⁷ prepared polydimethylsiloxane (PDMS) through a polycondensation reaction of dimethyldimethoxysilane and then introduced it into castor-oil-based waterborne polyurethane through a blending method, resulting in the formation of a modified PDMS castor-oil-based waterborne polyurethane coating. It was ascertained that the incorporation of PDMS into the coating led to the attainment of antifouling

properties, enhanced hydrophobicity, and increased water resistance while also enhancing tensile strength up to 5.2 MPa. In comparison to other bio-based polyurethane coatings, castor oil polyurethane (CO-PU) coatings exhibit a significantly faster drying rate. For instance, linseed oil polyurethane coatings may require several days to fully cure, limiting their practical application in industrial settings. In this study, an interpenetrating network (IPN)-structured polyurethane coating was developed, demonstrating superior hardness and wear resistance. While other bio-based coatings, such as tung-oil-based coatings, also exhibit high hardness, they suffer from poor flexibility, making them more susceptible to damage.

The fundamental element of marine antifouling coatings is the antifouling agent.^{18,19} Quaternary ammonium salts (QASs) have been verified to be highly efficacious and to manifest broad-spectrum antimicrobial activity against both Gram-negative and Gram-positive bacteria.^{20,21} Wang et al.²² modified the imidazole moiety on the side chain to obtain a polyurethane coating composed of a QAS and a zwitterion. It was ascertained that the coating was capable of generating a hydrated layer and exposing hydrophilic/hydrophobic groups on the surface, thereby fabricating micro/nanostructures that were lethal to bacteria and prevented the attachment of proteins and algae.

In this study, CO-PU was first synthesized. Subsequently, QASs were incorporated into CO-PU via blending to prepare QAS-modified CO-PU coatings. The effects of the QAS content and other factors on the mechanical properties, antibacterial performance, and antifouling efficacy of the coatings were systematically investigated. The results demonstrated that the QAS-modified coatings exhibited superior antifouling performance.

EXPERIMENTAL SECTION

Materials. Xylene, acrylonitrile, and styrene were procured from Sinopharm Group Chemical Reagent Co., Ltd. Diphenylmethane diisocyanate (MDI) was supplied by Wanhua Chemical Group Co., Ltd. Benzoyl peroxide (BPO) was obtained from Shanghai McLean Biochemical Technology Co., Ltd. Castor oil (CO) was procured from Inner Mongolia Weiyu Biotechnology Co., Ltd. E-44 epoxy resin was purchased from Henan Water Ring Industrial Co., Ltd. *Pseudomonas aeruginosa* was purchased from Beijing Preservation Biotechnology Co., Ltd.

Preparation of Coatings. The preparation procedures of QD-BC and PQDBCAM are elaborated in other literature.^{23,24} The primary synthesis scheme is depicted in Figure 1.

The average molecular weight of the synthesized QD-BC was 261, and that of PQDBCAM was 3064. After testing, it was found that both QASs exhibit good solubility in water but limited solubility in organic solvents. Both compounds demonstrated significant inhibitory effects against *Escherichia coli* and *Staphylococcus aureus*.

The following is an exemplification of the protocol for the circle of inhibition experiment for *E. coli*:

Weigh 0.05 g of QD-BC and PQDBCAM separately and dissolve each in 50 mL of sterile water to prepare 1 mg/mL solutions. Subject the solutions to ultrasonic treatment for 15 min. Add 200 μ L of bacterial suspension to 200 mL of sterilized solid medium, pour into a sterile Petri dish, and place an Oxford cup on the surface. Allow the medium to solidify before injecting 20 μ L of the QAS solution into each well. Incubate the plates in a constant temperature and humidity chamber for 12 h. As illustrated in Figure 2, both QD-BC and PQDBCAM exhibit antibacterial activity, with PQDBCAM demonstrating a larger inhibition zone diameter and superior antimicrobial efficacy.

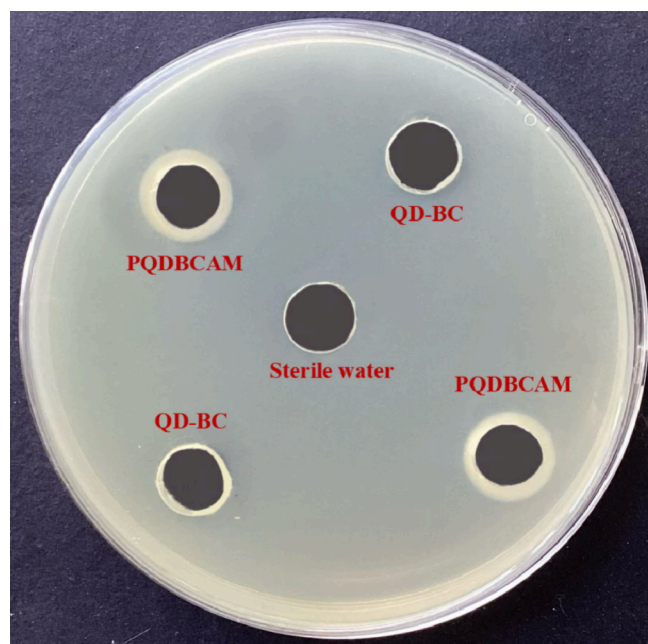


Figure 2. Circle of inhibition diagram of QASs.

Castor oil and xylene underwent a dehydration pretreatment. The experiment was conducted with a molar ratio of $-NCO$ to $-OH$ at 2.5:1. Dehydrated xylene (comprising approximately 25% of the total reactant mass) and diphenylmethane diisocyanate (MDI) were added to a three-necked flask, heated to 50 $^{\circ}C$, and stirred gently for 10 min. Subsequently, dehydrated castor oil was slowly dripped in, and the temperature was increased to 80 $^{\circ}C$ for a 2 h reaction period. Following this, dibenzoyl peroxide solution was introduced and thoroughly mixed, and the mixture underwent vacuum dehydration for 1 h, resulting in a golden yellow liquid castor oil polyurethane prepolymer.

Weigh 6 g of E-44 epoxy resin, 4 g of styrene, and 10 g of acrylonitrile and mix them thoroughly to obtain a homogeneous dispersion. Set this dispersion aside for later use. Next, prepare mixtures containing varying amounts of QASs (expressed as a percentage of the total mass of the dispersion and CO-PU), along with 1.4 g of the dispersion and 5 g of CO-PU. Mix these components uniformly and sonicate for 15 min. Subsequently, apply the mixture evenly onto a 5 $\times$ 5 cm iron plate and place it in a dust-free environment. Cure the coated plate at room temperature for 24 h,

followed by curing in an 80 $^{\circ}C$ oven for an additional 24 h. This process yields a CO-PU marine antifouling coating containing acrylamide-based QASs.

Mechanical Properties of Coatings. *Adhesion Test of Coatings.* The adhesion of the coating was appraised by utilization of a pull-off adhesion tester (Cangzhou Europol Testing Instrument Co., Ltd.), in consonance with the standards prescribed in ASTM D4541-02. Each sample is required to be tested in three distinct locations.

Pencil Hardness Test of Coatings. A pencil hardness tester (Cangzhou Europol Testing Instrument Co., Ltd.) was utilized for the assessment of the hardness of the coating film in consonance with the procedures delineated in GB/T6739-2006, the criterion for pencil hardness testing of color paint and varnish.

Flexibility Testing of Coatings. A paint film bending tester (Tianjin Jinghai County Kexin Testing Machine Factory Co., Ltd.) was utilized to determine the flexibility of the fabricated coatings in line with the standards stipulated in GB/T 1731-2020, "Determination of Flexibility of Paint Film and Putty Film".

Impact Strength Testing of Coatings. The impact resistance of the paint film was evaluated by means of a paint film impact tester (Cangzhou Europol Testing Instrument Co., Ltd.) in compliance with the standard GB/T 1732-2020, "Determination of Impact Resistance of Paint Film". A minimum distance of 15 mm was maintained at each drop point.

Thermal Stability Testing of Coatings. The coatings were segmented into blocks with a maximum diameter of 3 mm. The specimens were heated from room temperature to 800 $^{\circ}C$ at a rate of 10 $^{\circ}C/min$ under a nitrogen atmosphere. The thermal stability of the antifouling coatings was examined using a thermogravimetric analysis (TGA) model STA 449 F5 from Netzsch, Germany.

Water Contact Angle Testing of Coatings. A SCI3000E water contact angle meter was utilized to determine the water contact angle on the surface of the antifouling coating with the aim of exploring the hydrophilic/hydrophobic nature of the said coating.

Static Seawater Hydrolysis Testing. Samples were fabricated with diverse concentrations of QASs, with three samples in each group and dimensions of 5 $\times$ 5 cm. The initial weight m_0 was subsequently recorded. The samples were immersed in seawater at room temperature and sealed with plastic wrap. The experimental apparatus was placed in a light incubator maintained at 26 $^{\circ}C$ for cultivation under a photoperiod of 12 h of light and 12 h of darkness. The samples were extracted at 5 day intervals and rinsed gently with deionized water. They were then subjected to a 12 h vacuum drying process, during which they were kept at a constant weight. Subsequently, the sample was removed and weighed again, with the mass of each weighing noted as m_t ($t = 1, 2, 3, 4$, and 5). The seawater was refreshed every 5 days, and the sample data for each set of concentrations were averaged 3 times. The weight loss rate (M) of the samples was then acquired through calculation.²⁵

$$M = \frac{m_t - m_0}{m_0} \times 100\%$$

Antimicrobial Performance Testing of Marine Antifouling Coatings. The objective of this study was to assess the antimicrobial properties of marine antifouling coatings with *P. aeruginosa* as the test organism.

A volume of 400 μ L of the bacterial suspension, with a concentration of 5×10^5 , was applied on the coated surface and covered with a sterilized polyethylene film. The sample was then incubated in a constant temperature and humidity chamber at 37 $^{\circ}C$ for 12 h. Subsequently, the plate was soaked with 20 mL of sterile water for 1 h. Thereafter, 400 μ L was taken from the sterilized plate and spread using a spreading stick. Finally, the plate was inverted and incubated in a constant temperature and humidity oven for 12 h. The antimicrobial rate W was determined by taking photographs and counting the number of colonies. The antimicrobial rate (W) was calculated as the number of colonies present on the plate.

$$W = \frac{W_0 - W_1}{W_0} \times 100\%$$

Table 1. Formulation of Sample Solutions

Centrifuge tube number	1	2	3	4	5	6	0%-1	0%-2	0%-3	1%-1	...
Standard protein BSA (μL)	0	40	80	120	160	200	200	200	200	200	...
PBS (μL)	200	160	120	80	40	0	0	0	0	0	0
BCA working fluids (mL)	2	2	2	2	2	2	2	2	2	2	2

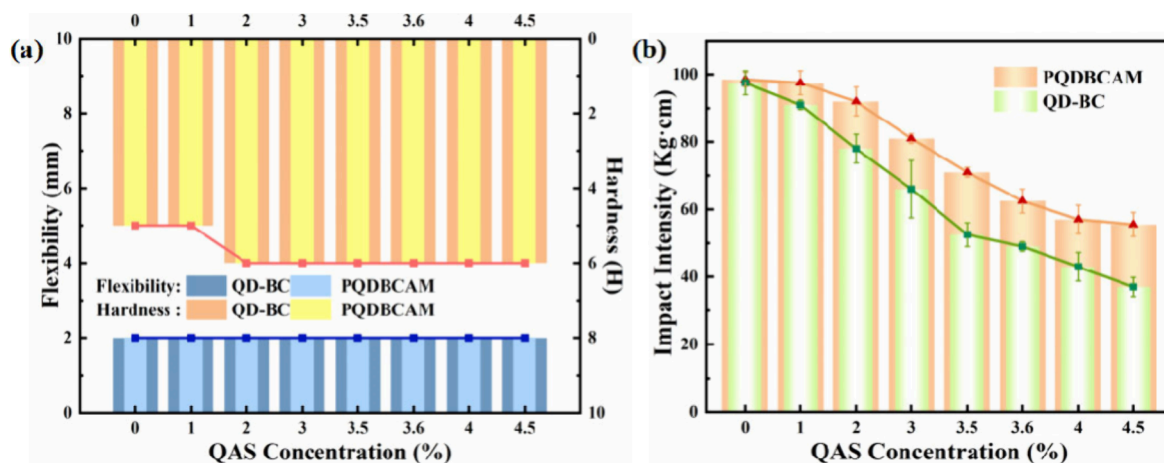


Figure 3. Mechanical properties of coatings.

In this experiment, W_0 denotes the number of bacteria in the blank control group, while W_1 denotes the number of bacteria in the experimental group.

Antimicrobial Durability Testing of Marine Antifouling Coatings. The coatings were subsequently examined for antimicrobial properties following a 3 month period of storage, in line with the aforesaid conditions.

Diatom Attachment Experiment of Marine Antifouling Coatings. Experiments were carried out to evaluate the adhesion ability of diatoms to the coatings.

Cultivation approaches for diatoms:

- (1) The filtered seawater should be transferred to a 1 L conical flask and sealed with an appropriate sealing film. The flask should then be sterilized by autoclaving, after which it should be cooled to ambient temperature.
- (2) On the ultraclean bench, the appropriate amount of diatom seeds is to be placed in conical flasks. The appropriate amount of nutrient solution is to be added, and the flasks are then to be placed in an environment maintained at 26 °C for the purpose of cultivation (12 h of light/12 h of darkness).

The prepared coating was placed in a beaker, and 20 mL of a diatom suspension of a specific concentration was taken to submerge the coating. The beaker was then incubated for 7 days at 26 °C (12 h of light and 12 h of darkness). Subsequently, the coating was gently rinsed with deionized water to remove any diatoms that were not firmly adhered to the surface. The diatom attachment was then observed and photographed for documentation using a microscope.

Protein Adhesion Resistance of Marine Antifouling Coatings. To assess the adhesion capacity of the coating to bovine serum albumin (BSA), protein adsorption experiments were carried out using the BCA protein kit method, as referenced in the literature.^{26,27}

- (1) The coated side of the sample to be tested (5×5 cm) was shielded with tin foil, and the shielded sample was equilibrated in phosphate-buffered saline (PBS) buffer solution (pH 7.4) for 1 h. The sample was then placed in the prepared 2 mg/mL BSA/PBS buffer and incubated at room temperature for 2 h.
- (2) Remove the aluminum foil and wash it with PBS buffer. Take the center position and cut out three small samples of 1×1 cm. Put them into a centrifuge tube containing 2 mL of PBS buffer and number them as 0%-1, 0%-2, 0%-3, 1%-1, ... Finally, conduct ultrasonic treatment for 30 min.

- (3) Take 40 μL of the wash solution and dilute it 10-fold with 360 μL of PBS buffer.
- (4) Take 100 μL of BCA standard and add it to a 1 mL PBS buffer solution at a concentration of 0.5 mg/mL. Subsequently, add the reagents in accordance with the specifications described in Table 1 and plot the standard curve.
- (5) Take 200 μL of the diluted sample (step 3) and 2 mL of the BCA working solution, mix them thoroughly, and then employ an ultraviolet–visible (UV–vis) spectrophotometer to measure the absorbance at 562 nm. The concentration of the protein solution to be tested can subsequently be determined using the standard curve. The mean value was calculated for each group.

$$\text{The quantity of BSA adsorbed per unit area} = \frac{10CV}{S}$$

The unit is $\mu\text{g}/\text{cm}^2$, where C represents the measured protein concentration ($\mu\text{g}/\text{mL}$), V indicates the volume of the PBS solution in the wash solution (mL), and S denotes the surface area (cm^2).

RESULTS AND DISCUSSION

Mechanical Properties of Coatings. As depicted in Figure 3b, the impact resistance manifested a decline with an increase in QAS content, and the QD-BC antifouling coating presented a more notable reduction. As depicted in Figure 3a, the pencil hardness and flexibility of both the QD-BC and PQDBCAM antifouling coatings display a similar trend. The flexibility is 2 mm, and the pencil hardness increases from 5H to 6H.

This phenomenon can be attributed to the benzene ring structure in the MDI molecule, which imparts a higher rigidity to the polyurethane resin molecular chain segments. Additionally, the CO-PU prepolymer undergoes cross-linking during the curing process to form a three-dimensional (3D) network, thereby enhancing the mechanical strength of the coating. The rigid benzene ring structure of the QASs (Figure 4) enhances the pencil hardness of the coating. This leads to an overall harder coating, while its brittleness also increases, making it susceptible to fracture when exposed to external forces. Consequently, its impact resistance also decreases. In contrast,

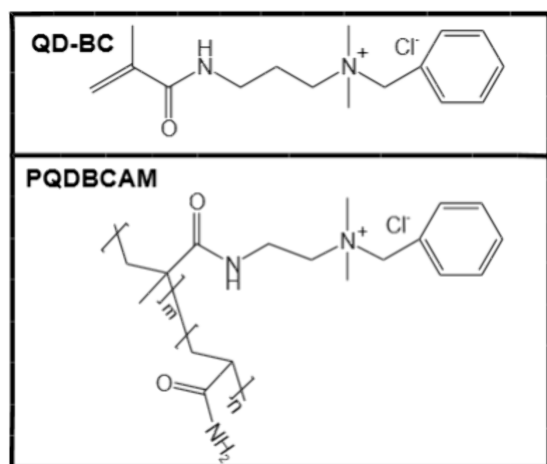


Figure 4. Structural formula of QAs in coatings.

PQDBCAM antifouling coatings are capable of resisting the impact of a height of 60 cm, which indicates that the PQDBCAM antifouling coatings exhibit superior impact resistance compared to others. The superior impact resistance of these coatings can be attributed to the reactive groups, such as amino groups, in the PQDBCAM molecular structure, which can react with the isocyanate groups in the polyurethane prepolymer. This reaction increases the number of cross-linking points and forms a 3D network structure, thereby enhancing the coating's ability to withstand greater impact forces. Consequently, the impact resistance of the PQDBCAM antifouling coating surpasses that of other coatings.

As seen in Figure 5, the adhesion of the coatings exceeds 1 MPa, which is attributed to the existence of functional groups,

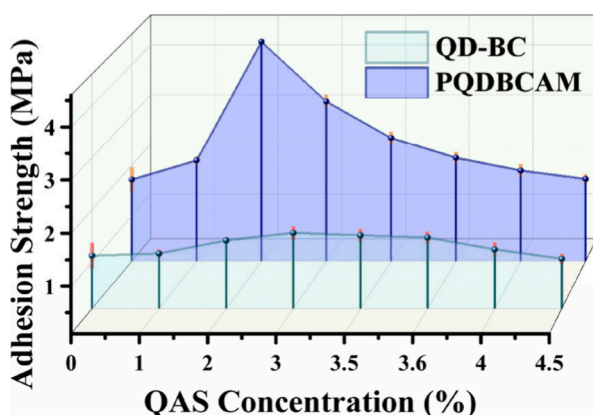


Figure 5. Adhesion of coatings.

such as hydroxyl and ester groups, within the coatings. These functional groups are capable of interacting with atoms or molecules on the surface of the substrate, thereby strengthening the adhesion. The adhesion of the PQDBCAM antifouling coating rises initially and then declines. The maximum adhesion of the PQDBCAM coating is 4.79 MPa. This can be attributed to the enhanced cationic strength of the polymer QAs, which leads to a more superior adhesion in comparison to that witnessed with the monomer. Conversely, an increase in the concentration of QAs resulted in a decrease in adhesion. This phenomenon can be ascribed to the propensity of the QAs to absorb water when the coating is indoors. An increase in the concentration of the QAs leads to a

corresponding increase in the water absorption, thereby ultimately causing a decrease in adhesion.

Thermal Stability Testing of Coatings. As seen in Figure 6, the thermal degradation process of the coating can be delineated into three distinct phases.²⁸ The initial stage occurs within the temperature range of 200–350 °C. The coatings containing QAs exhibited minor peaks in the DTG curves, while no peaks were observed in the pure coatings. This peak can be ascribed to the rupture of molecular chains within the QAs, which emerges as a thermal decomposition peak. As demonstrated by Figure 6, the thermal decomposition temperature of the QAs in the PQDBCAM coating is higher than that of the QAs in the QD-BC coating.

The second stage ranges from 350 to 400 °C with a mass loss of approximately 21%. This phenomenon might be attributed to the degradation of hard segments of urethane into small molecular chain segments or volatile small-molecule gases. The third stage is characterized by a temperature range of 400–500 °C and a mass loss ranging from 42 to 48%. This might signify a degradation process taking place within the polyester chain segments, resulting in the transformation of the soft segments into chain hydrocarbons and carbon dioxide. The maximum weight loss rate was observed to be 459.2 °C for the pure coating, 461.6 °C for the 4.5% QD-BC coating, and 459.0 °C for the 3.6% PQDBCAM coating. These findings suggest that thermal stability was enhanced by the incorporation of the QAs.

Water Contact Angle of Coatings. As depicted in Figure 7, the water contact angle of both coatings is less than 90°, indicating that their surfaces exhibit hydrophilic properties. The hydroxyl groups and free isocyanate in CO-PU resin on the surface of the coating and the amino and hydroxyl groups in the QAs are all hydrophilic groups that are capable of forming hydrogen bonds with water molecules. Consequently, these groups enable the coating surface to adsorb water molecules, resulting in hydrophilicity. However, the water contact angle of the PQDBCAM coating is larger than that of the QD-BC coating. This difference arises because, during the coating preparation process, some hydrophilic groups become entangled within the coating matrix, while those exposed on the surface contribute to a slight increase in surface roughness. As a result, the overall hydrophilicity of the PQDBCAM coating is reduced, leading to a higher water contact angle.

Static Seawater Hydrolysis Test. The marine antifouling coatings were submerged in seawater for 30 days, and their weights were measured at 5 day intervals. The obtained data were plotted on a graph of the weight loss rate versus time to facilitate the investigation of the hydrolysis of the coatings under static seawater conditions. As shown in Figure 8, the weight loss rate of both the QD-BC marine antifouling coating and the PQDBCAM marine antifouling coating is less than 0.5%. The weight loss rate of the coatings was initially higher during the first 10 days of immersion, followed by a flattening of the weight loss rate curves, suggesting stabilization of the coating quality. This indicates that the coating exhibits enhanced water resistance.

Antimicrobial Performance Testing of Coatings. As depicted in Figures 9 and 10, both QD-BC and PQDBCAM marine antifouling coatings showed effectiveness in inhibiting the growth of *P. aeruginosa*. This phenomenon can be attributed to the fact that QAS groups carry a positive charge, whereas the surfaces of bacterial cells possess a negative charge. The positively charged coating surface can effectively inhibit

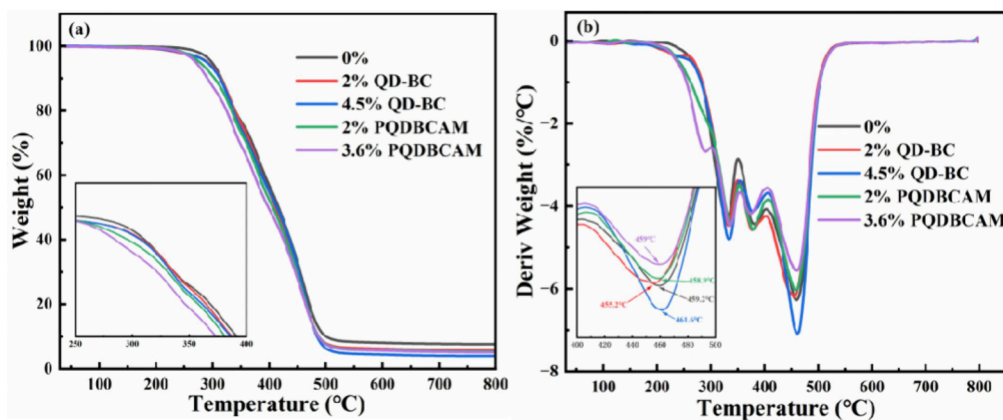


Figure 6. Thermogravimetric analysis of coatings: (a) thermogravimetry (TG) and (b) differential thermogravimetry (DTG) graphs.

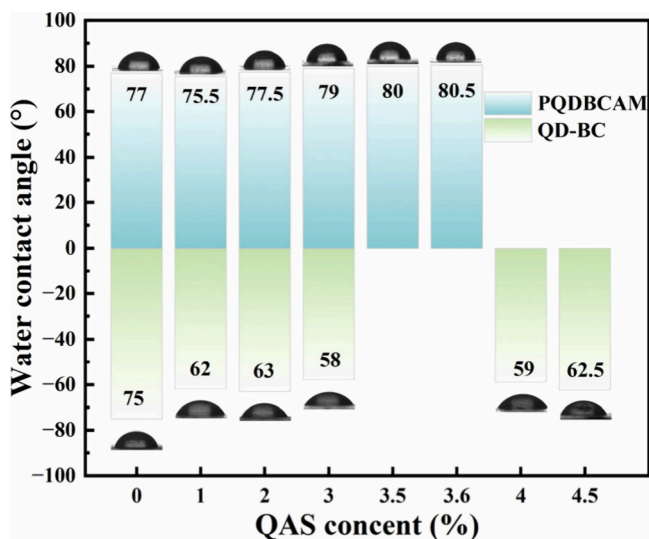


Figure 7. Water contact angle of coatings.

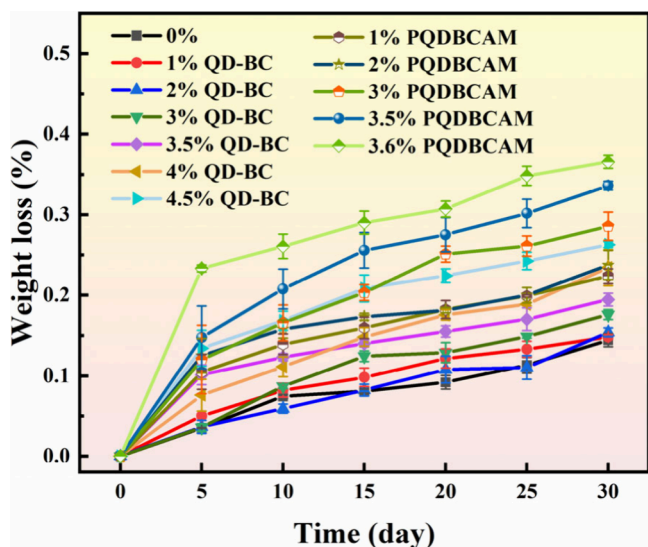


Figure 8. Weight loss of the coatings.

the attachment of fouling organisms through electrostatic repulsion. The combination of PQDBCAM and QD-BC exhibited an inhibitory effect against *P. aeruginosa* exceeding 99% at a ratio of 3.6% PQDBCAM and 4.5% QD-BC.

Additionally, the antimicrobial efficacy of the PQDBCAM antifoulant surpassed that of the QD-BC marine antifoulant at equivalent concentrations, indicating a superior antimicrobial capacity of the polymers compared to the corresponding monomers. This is because monomer polymerization, an increase in density, and enhanced cell attraction can change the permeability of the cell, damage the cell membrane, and ultimately cause the leakage of intracellular substances, which, in turn, leads to cell death. This process is known to have an antibacterial effect.

Antimicrobial Durability Testing of Coatings. As shown in Figure 11, the number of bacteria adhering to plates without QASs is significantly higher. As the concentration of QASs increases, the bacterial count on the plates gradually decreases, with a more rapid reduction observed in the samples containing PQDBCAM. Figure 10 indicates that, after 3 months, although the antibacterial efficacy of the coating slightly decreases compared to its initial state, it still exhibits significant inhibitory effects against *P. aeruginosa*. Specifically, when the PQDBCAM content is 3.6% and the QD-BC content is 4.5%, the antibacterial rate remains at 99%, demonstrating the coating's excellent long-term antibacterial durability.

Evaluation of Algae Inhibition Properties of Coatings. Figure 12 shows the surface of the antifouling coating after a 7 day immersion in the diatom culture solution, with Figure 12e presenting the diatom coverage on the aforesaid coating. The data indicated that 0% of the coatings (panels a and c of Figure 12) showed a high degree of diatom attachment, whereas only minimal diatom attachment was observed on the surfaces of QD-BC-4.5% (Figure 12b) and PQDBCAM-3.6% (Figure 12d) coatings, with an area of algal coverage of approximately 22.3%. Experimental results reveal that the addition of hydroxyl groups to a QAS leads to a significant improvement in the coating's hydrophilicity, which can cause a considerable reduction in the growth and spread of algae colonies on the surface of the coating. This is further exemplified by the remarkable decrease in the area occupied by algae, particularly in the PQDBCAM antifouling coating.

This phenomenon can be attributed to the hydrophilic property of the coating, which promotes the formation of a hydration layer on the surface. Another reason is that, when algal cells approach the coating surface, electrostatic interactions make it difficult for the algae to adhere and proliferate on the surface, thereby enhancing the antifouling performance of the coating. This layer effectively inhibits direct contact between the diatoms and the coating surface. The intense

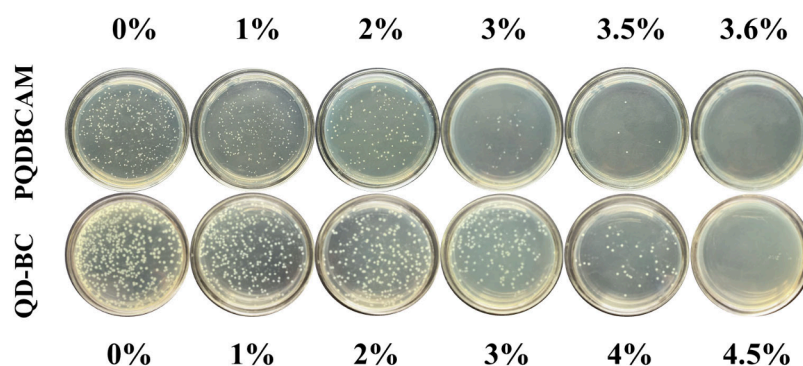


Figure 9. Antibacterial efficacy of QD-BC and PQDBCAM marine antifouling coating against *P. aeruginosa*.

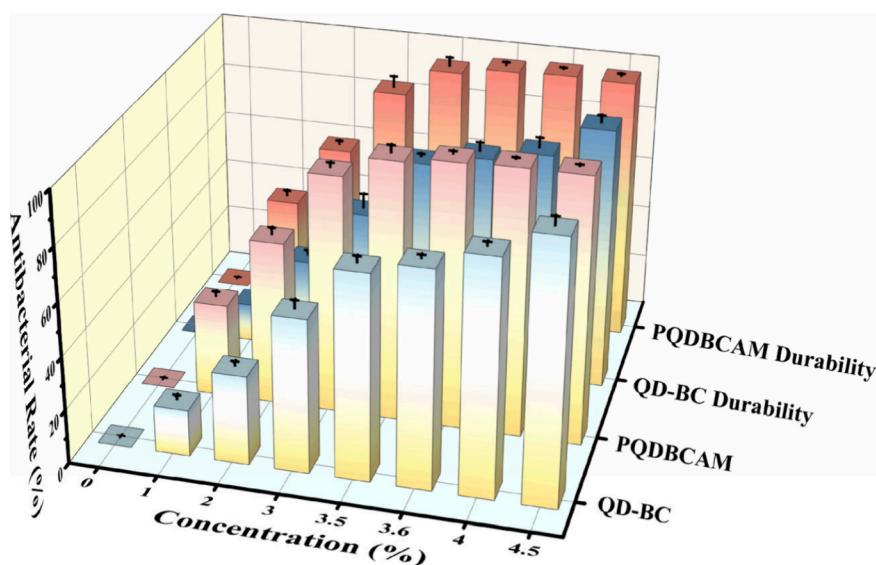


Figure 10. Antimicrobial rate of QD-BC and PQDBCAM marine antifouling coating against *P. aeruginosa*.

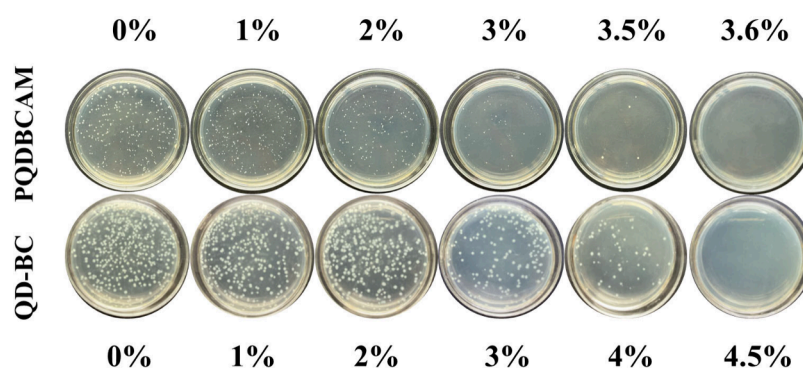


Figure 11. Antibacterial durability test of QD-BC and PQDBCAM marine antifouling coating against *P. aeruginosa*.

hydration of the polymer QASs gives rise to a more conspicuous inhibition effect of the PQDBCAM coating on the diatoms. However, when the content of the QASs is excessive, uneven local dispersion might lead to a differential inhibition effect on diatoms.

Evaluation of Coatings for Protein Adhesion Resistance. The protein BSA standard curve was drawn by testing the absorbance of the standard protein solution at 562 nm, as shown in Figure 13a. The absorbance of the coating was quantified, and the protein concentration was calculated from the standard curve presented in Figure 13b. An increase in the concentration of QASs led to a decrease in protein adsorption

for all of the coatings. This was attributed to the formation of a hydration layer on the surface of the coatings, which hindered the adsorption of non-specific proteins and enabled them to display stable anti-protein adsorption properties. The BSA adsorption of the PQDBCAM antifouling coating shows a lower level of interaction compared to that of the QD-BC antifouling coating. This can be ascribed to the enhanced affinity of the polymer macromolecules for water, which allows the formation of a more compact hydration layer and, consequently, a more effective antifouling coating.

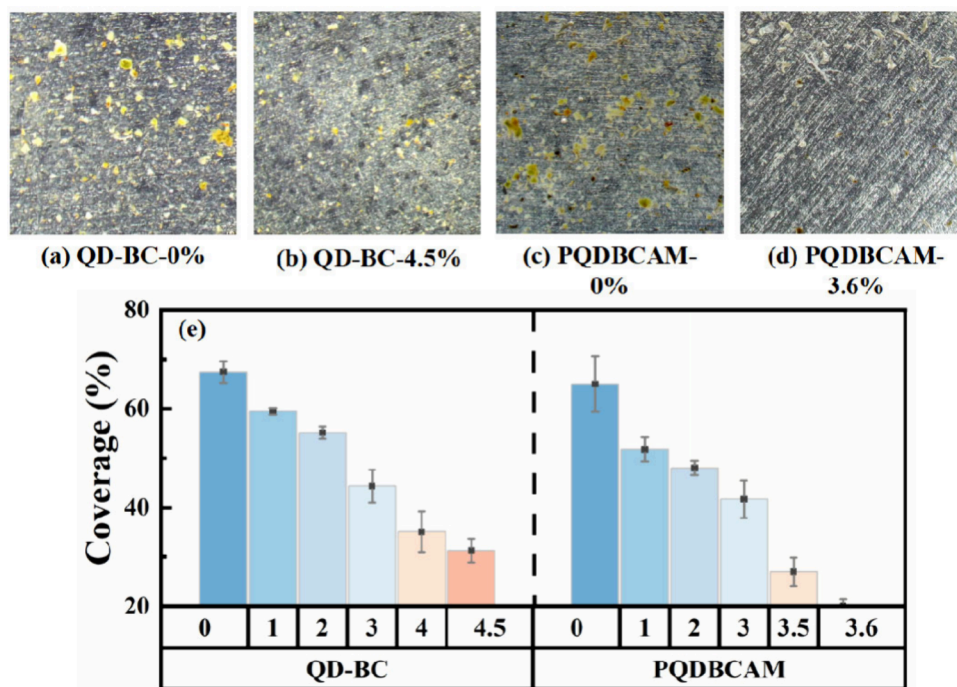


Figure 12. Diatom attachment.

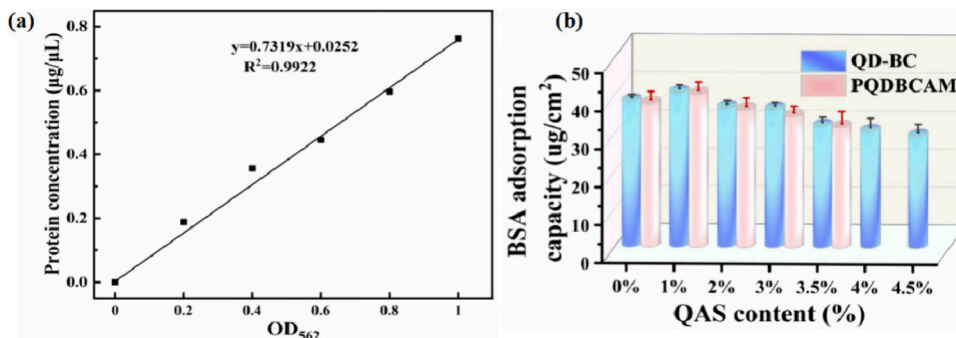


Figure 13. (a) Standard protein curve and (b) protein adsorption.

CONCLUSION

This study primarily focuses on the development of environmentally friendly marine antifouling coatings. Castor oil, a bio-based polyol, is utilized to replace petroleum-derived raw materials and is reacted with MDI to synthesize bio-based polyurethane resin, thereby reducing environmental pollution. The acrylamide QASs QD-BC and PQDBCAM were incorporated into castor-oil-based polyurethanes to obtain the resultant QD-BC and PQDBCAM castor-oil-based polyurethane emulsion and coatings. The addition of the acrylamide QASs brought about an improvement in the hydrophilic properties of the coating. Nevertheless, the adhesion of the PQDBCAM coating was discovered to be superior to that of the QD-BC coating, with a maximum adhesion force of 4.79 MPa. The temperature at which the maximum rate of coating weight loss is observed is approximately 460 °C. All of the coatings display good thermal stability. The weight loss rates of the coatings were all less than 0.5%, indicating a negligible inhibitory effect on *P. aeruginosa*. Among the tested coatings, the polymer QASs showed the greatest efficacy, achieving over 99% inhibition of *P. aeruginosa* at a PQDBCAM of 3.6%. All

coatings exerted inhibitory effects on diatom and protein growth. The polymer QASs were found to be more effective.

Despite the environmental friendliness of the coatings studied in this work, there is a possibility of QAS leakage into the marine environment in minimal amounts, which could potentially affect the marine ecosystem. Physical or chemical methods have been adopted to better fix polyacrylamide QASs in the coating to address this potential issue. According to different application scenarios, the formula and construction process of the paint are adjusted to achieve the application purpose.

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Notes

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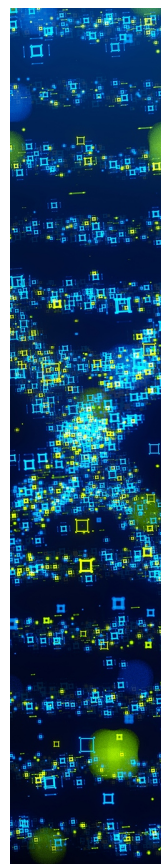
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